$$\sum_{r \geq 0} R(p^r) X^r = \frac{1 - \chi_5(p) p^{10} X^2}{1 - \tau(p) X + p^{11} X^2}$$

Numerator removes the square-content contribution

$$R(p^r) = \tau(p^r) - \chi_5(p) p^{10} \tau(p^{r-2})$$

for r at least 2

Formal series in X at one fixed prime p.